

LA ROCHELLE LE BOUT BLANC, France

WMO: 073150

Lat: 46.1526N

Lon: 1.1587W

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	-0.9	-6.5	2.2	-0.8	-4.7	2.6	0.7	13.2	11.1	11.5	11.1	3.4	80	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	30.0	20.1	27.7	19.6	25.9	19.1	21.9	27.0	21.0	25.6	20.2	24.2	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.0	24.1	19.4	14.2	23.1	18.7	13.6	22.3	63.9	27.2	60.7	25.7	58.0	24.4	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.3	8.1	DB	-4.2	34.3	2.1	2.2	-5.8	35.9	-7.0	37.2	-8.2	38.5	-9.7	40.1	(4)
			WB	-5.0	23.4	2.1	1.0	-6.5	24.1	-7.8	24.7	-8.9	25.3	-10.4	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	6.8	7.4	9.9	12.1	15.5	18.7	20.3	20.3	17.9	15.0	10.5	7.3	(6)
		DBStd	5.75	3.60	3.28	2.92	2.75	2.79	2.79	2.66	2.45	2.52	2.94	3.23	3.80	(7)
		HDD10.0	363	105	79	37	10	0	0	0	0	3	32	97		(8)
		HDD18.3	1973	357	306	261	188	98	27	8	6	37	109	235	341	(9)
		CDD10.0	1644	7	7	34	72	169	261	319	321	237	157	47	14	(10)
		CDD18.3	213	0	0	0	1	9	38	69	68	23	4	0	0	(11)
		CDH23.3	1137	0	0	0	5	65	236	377	328	117	9	0	0	(12)
(14)	Wind	WSAvg	4.0	4.0	4.2	4.2	4.3	4.2	4.1	4.2	3.9	3.6	3.4	3.9	4.0	(14)
		PrecAvg	735	74	56	50	61	51	36	44	47	56	83	96	80	(15)
		PrecMax	1016	166	118	162	240	88	85	130	109	210	173	259	148	(16)
		PrecMin	524	10	0	5	15	15	6	5	0	10	18	14		(17)
		PrecStd	131	40	32	36	45	21	18	23	32	47	42	50	38	(18)
		DB	13.7	15.3	21.1	23.9	27.9	32.2	32.7	32.2	29.7	24.6	18.7	14.7		(19)
		MCWB	11.7	11.0	13.3	15.9	18.1	20.5	21.5	21.3	19.1	18.0	15.1	13.0		(20)
(22)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	12.7	13.3	18.1	20.6	25.2	28.1	29.8	29.3	26.2	22.0	16.7	13.7		(21)
		MCWB	11.3	10.4	12.3	14.0	17.5	19.3	20.1	20.6	18.4	17.4	14.3	12.2		(22)
		DB	12.0	12.2	15.9	18.8	22.6	25.7	27.4	26.9	24.2	20.3	15.7	13.0		(23)
		MCWB	10.7	10.3	11.4	13.2	16.6	18.8	19.6	20.2	17.7	16.8	13.9	11.7		(24)
		DB	11.3	11.4	14.1	16.9	20.4	23.8	25.1	24.9	22.4	19.0	14.8	12.2		(25)
		MCWB	10.1	9.9	10.7	12.5	15.7	18.0	19.2	19.7	17.5	16.3	13.1	11.1		(26)
		WB	12.5	12.2	14.2	16.6	20.1	22.2	22.5	23.8	21.3	19.4	16.0	13.6		(27)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	13.1	13.5	19.2	21.2	25.9	28.7	28.4	28.6	26.0	22.1	17.4	14.3		(28)
		WB	11.6	11.4	13.0	15.1	18.2	20.9	21.5	22.2	20.1	18.3	15.1	12.6		(29)
		MCDB	12.4	12.5	16.5	19.1	23.2	26.2	26.9	26.6	23.5	20.6	16.2	13.4		(30)
		WB	11.0	10.7	12.2	13.9	17.1	19.8	20.7	21.2	19.2	17.5	14.2	11.9		(31)
		MCDB	11.7	11.7	14.6	17.5	21.4	24.3	25.2	25.2	22.3	19.6	15.4	12.7		(32)
		WB	10.3	10.2	11.5	12.9	16.1	18.7	19.9	20.2	18.3	16.7	13.3	11.2		(33)
		MCDB	11.2	11.3	13.4	15.9	19.7	22.6	24.0	23.6	21.4	18.7	14.7	12.2		(34)
(35)	Mean Daily Temperature Range	MDBR	4.7	5.3	6.5	6.9	6.6	7.0	6.5	6.8	7.5	6.4	5.3	4.9		(35)
		5% DB	4.3	6.2	9.9	10.7	11.4	11.6	11.6	11.1	11.1	8.2	5.9	4.5		(36)
		MCWBR	3.6	4.4	5.8	5.9	5.7	5.4	5.2	5.3	5.5	4.8	4.1	3.8		(37)
		5% WB	3.9	4.8	7.3	8.9	9.3	9.6	8.8	8.4	8.1	6.6	5.2	4.0		(38)
		MCWBR	3.5	3.8	4.7	5.4	5.2	5.1	4.6	4.7	4.9	4.2	4.1	3.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.344	0.373	0.394	0.399	0.410	0.401	0.392	0.370	0.368	0.342	0.326		(40)
		taud	2.459	2.411	2.336	2.282	2.323	2.323	2.348	2.391	2.429	2.440	2.450	2.444		(41)
		Ebn at Noon	766	826	850	862	869	858	861	856	845	788	741	724		(42)
		Edh at Noon	69	87	108	124	124	124	120	110	98	84	69	64		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.21	2.17	3.35	4.75	5.57	6.24	6.03	5.34	4.24	2.54	1.47	1.07		(44)
		RadStd	0.12	0.22	0.37	0.45	0.43	0.48	0.29	0.38	0.28	0.22	0.16	0.13		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page